Cox-Process Scenario Generation, Decomposition, and Distributional Robustness for CVaR-Based Stochastic Reshoring

Extension of the Rockafellar–Uryasev linearization framework
applied to the BOM-DAG decision model of Chandran & Shanmuganathan (2026)

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Abstract

The CVaR-based reshoring framework substitutes the Rockafellar–Uryasev variational form of Conditional Value-at-Risk for the variance penalty in Chandran & Shanmuganathan's (2026) mean-plus-variance objective, preserving linearity of the decision program. The substitution is mathematically clean but inherits a silent assumption: that scenario realizations of route, tariff, and assembly costs are drawn independently across ω. Empirically, supply-chain disruptions cluster—driven by shared geopolitical, climatic, and macroeconomic intensity. We propose a Cox-process generator for the cost coefficients, calibrated to a latent affine intensity λ_t, which produces tail scenarios with realistic positive cross-route correlation. We further develop four optimization-side improvements: stratified sampling on intensity quantiles, Benders decomposition exploiting the LP structure of the CVaR constraints, generalization to spectral risk measures, and Wasserstein-distributionally-robust CVaR on the intensity parameters. A worked three-route example demonstrates that under the Cox generator the optimal sourcing strategy genuinely diversifies, whereas under independent sampling the cheapest single route dominates. We close with a protocol for out-of-sample validation under tail-correlated scenarios.

Contents

1 Introduction **2**
1.1 Motivation . . . 2
1.2 Contribution . . . 3

2 Recapitulation of the Decision Model **3**
2.1 BOM-DAG and decision variables . . . 3
2.2 Constraints . . . 3
2.3 Stochastic landed cost . . . 4
2.4 Mean-plus-CVaR objective . . . 4

3 Cox-Process Scenario Generator **4**
3.1 Latent intensity dynamics . . . 4
3.2 Doubly stochastic counting process . . . 4
3.3 Coupling cost coefficients to the intensity . . . 5
3.4 Tail dependence: independent vs. Cox . . . 5

4 Stratified Sampling on Intensity Quantiles **5**
4.1 The variance problem . . . 5
4.2 Stratification scheme . . . 6

5 Benders Decomposition of the CVaR-MILP 6
5.1 Block structure . . . 6
5.2 Cuts . . . 6
5.3 Convergence . . . 7

6 Spectral Risk Generalization 7
6.1 Definition . . . 7
6.2 LP reformulation . . . 7

7 Distributionally Robust CVaR 7
7.1 Motivation . . . 7
7.2 Wasserstein ambiguity . . . 8
7.3 Tractable reformulation . . . 8
7.4 Choice of radius . . . 8

8 Worked Example: Three-Route Component Sourcing 8
8.1 Setup . . . 8
8.2 Cox calibration . . . 9
8.3 Comparison: independent vs. Cox . . . 9
8.4 Visualization of cost distributions . . . 9
8.5 Diagnostic: tail correlation . . . 9

9 Out-of-Sample Validation Protocol 10
9.1 Why naive holdout is insufficient . . . 10
9.2 Procedure . . . 10
9.3 Recommended budgets . . . 10

10 Discussion 10
10.1 What changed and what did not . . . 10
10.2 Coherence and convexity, revisited . . . 10
10.3 Scope and limitations . . . 11
10.4 Practical recommendation . . . 11

A Proof sketch of Proposition 3.2 11

B Numerical recipe for the stratified estimator 11

1 Introduction

1.1 Motivation

The reshoring framework of Chandran & Shanmuganathan [3] formulates sourcing, routing, and assembly as a sample-average mixed-integer program over a Bill-of-Materials directed acyclic graph (BOM-DAG). Its original objective is mean-plus-variance, which the recent CVaR-based reformulation [4] replaces with mean-plus-CVaR via the variational form of Rockafellar & Uryasev [1]. The substitution achieves three goals simultaneously: it restores linearity (yielding a MILP rather than a MIQP), restores coherence in the sense of Artzner et al. [5], and orients the risk term toward downside loss rather than symmetric deviation.

What the substitution does *not* address is the structure of the scenario set itself. The cost coefficients $c_{ij}^{\text{route}}(\omega)$ and $c_{ik}^{\text{assy}}(\omega)$ in equation (10) of [4] are treated as exogenously given, and their scenario realizations are silently assumed to be statistically independent across (i, j, ω) and (k, ω). This is convenient for Monte Carlo but empirically wrong. The disruption events that motivate reshoring—trade conflict, port closure, sanctions waves, climatic shocks, pandemic surges—do not arrive as independent jitter on each route. They arrive as *co-shocks driven by a shared latent intensity*, with cross-route effect amplitudes determined by exposure.

When scenarios are independent, CVaR computes the tail expectation of statistical noise. The optimizer responds by selecting the cheapest single route, since tail co-occurrence across alternatives has vanishing probability. When scenarios share a latent intensity, the CVaR tail is populated by realistic co-shocks, and diversification across exposure profiles becomes Pareto-efficient. The risk measure does not change; the scenarios it measures do.

1.2 Contribution

This paper develops five extensions to the CVaR-based framework, none of which alter the decision variables or constraint structure of [3, 4]:

1. A **Cox-process scenario generator** (Section 3) coupling route, tariff, and assembly cost shocks to a shared affine intensity λ_t, with route-specific exposure κ_{ij}. Disruptions cluster, and tail scenarios exhibit realistic positive cross-route correlation.

2. A **stratified sampling** scheme (Section 4) on intensity quantiles, reducing CVaR estimator variance by an order of magnitude relative to crude Monte Carlo at fixed budget.

3. A **Benders decomposition** (Section 5) of the resulting MILP, exploiting the fact that for fixed binary decisions the CVaR subproblem is a pure linear program.

4. A **spectral risk generalization** (Section 6) replacing the single confidence level α with a weighting kernel ϕ over the quantile range, while preserving the LP reformulation.

5. A **distributionally robust CVaR** formulation (Section 7) over a Wasserstein ball around the calibrated Cox parameters, addressing the small-sample bias in CVaR estimation noted but not resolved in [4, 6].

A worked numerical example (Section 8) and an out-of-sample validation protocol (Section 9) close the development. Throughout, our principle is conservative: we change the inputs and the solution method, never the decision model.

2 Recapitulation of the Decision Model

We summarize the formulation of [3, 4] for self-containment.

2.1 BOM-DAG and decision variables

Components $i \in N = \{1, \dots, n\}$ form a directed acyclic graph $G = (N, A)$ of assembly dependencies. Each i carries attributes $a_i = (\tau_i, d_i, w_i, v_i, \ell_i)$: tariff classification, duty rate, weight, volume, labor effort. Decision variables:

$$z_i \in \{0,1\}, \qquad \text{(sourcing: 1 offshore, 0 domestic)} \tag{1}$$
$$r_{ij} \in \{0,1\}, \quad j \in R_i, \qquad \text{(route, applicable if } z_i = 1) \tag{2}$$
$$a_{ik} \in \{0,1\}, \quad k \in K, \qquad \text{(assembly location, applicable if } z_i = 0). \tag{3}$$

2.2 Constraints

$$\sum_{j \in R_i} r_{ij} = z_i, \qquad \sum_{k \in K} a_{ik} = 1 - z_i, \qquad \sum_{i \in N} \ell_i \, a_{ik} \le C_k \ \ \forall k, \qquad T^{\omega}(r, a) \le T_{\text{SLA}} \ \ \forall \omega. \tag{4}$$

2.3 Stochastic landed cost

For each scenario ω,

$$Z^{\omega}(z,r,a) = \sum_{i\in N}\left[\sum_{j\in R_i} r_{ij}\, c_{ij}^{\text{route}}(\omega) + \sum_{k\in K} a_{ik}\, c_{ik}^{\text{assy}}(\omega)\right]. \tag{5}$$

The functional form of $c_{ij}^{\text{route}}(\omega)$ and $c_{ik}^{\text{assy}}(\omega)$ is left exogenous in [3]; specifying it is the entry point for the Cox-process extension below.

2.4 Mean-plus-CVaR objective

Let $\alpha \in (0,1)$ be the confidence level. By the Rockafellar–Uryasev theorem [1],

$$\mathrm{CVaR}_{\alpha}[Z] = \min_{\zeta\in\mathbb{R}}\left\{\zeta + \frac{1}{(1-\alpha)\Omega}\sum_{\omega=1}^{\Omega}\max(0, Z^{\omega}-\zeta)\right\}. \tag{6}$$

With slacks $s_\omega \geq Z^\omega - \zeta$, $s_\omega \geq 0$, the program (22) of [4] becomes

$$\boxed{\min_{z,r,a,\zeta,s}\ \frac{1}{\Omega}\sum_{\omega=1}^{\Omega} Z^{\omega} + \lambda\left[\zeta + \frac{1}{(1-\alpha)\Omega}\sum_{\omega} s_\omega\right] \quad \text{s.t. constraints (4)–(7)},\ s_\omega \geq Z^\omega - \zeta,\ s_\omega \geq 0.} \tag{7}$$

This is a mixed-integer linear program for any specification of $c_{ij}^{\text{route}}(\omega)$ and $c_{ik}^{\text{assy}}(\omega)$ that is affine in the decisions for each ω.

3 Cox-Process Scenario Generator

3.1 Latent intensity dynamics

Let λ_t denote a latent disruption intensity following an affine (CIR-type) diffusion:

$$d\lambda_t = a(b-\lambda_t)\,dt + \eta\sqrt{\lambda_t}\,dW_t^{\lambda}, \quad \lambda_0 > 0, \tag{8}$$

with mean-reversion speed $a > 0$, long-run level $b > 0$, and volatility-of-volatility $\eta \geq 0$. The Feller condition $2ab \geq \eta^2$ ensures $\lambda_t > 0$ almost surely.

3.2 Doubly stochastic counting process

A Cox process N_t has $\mathbb{P}(dN_t = 1 \mid \mathcal{F}_t) = \lambda_t\,dt$, so the conditional rate of disruptions is itself stochastic and correlated across time. The marginal distribution of N_T is mixed-Poisson with random parameter $\Lambda_T = \int_0^T \lambda_s\,ds$:

$$\mathbb{P}(N_T = k) = \mathbb{E}\left[\frac{\Lambda_T^k e^{-\Lambda_T}}{k!}\right]. \tag{9}$$

Because Λ_T has positive variance, $\mathrm{Var}(N_T) > \mathbb{E}[N_T]$: the process is overdispersed relative to a homogeneous Poisson process. This overdispersion is precisely the empirical signature of clustered supply-chain disruptions.

3.3 Coupling cost coefficients to the intensity

We specify

$$c_{ij}^{\text{route}}(\omega) = c_{ij}^{\text{route,base}} + \kappa_{ij}^{\text{route}}\, N_T(\omega), \tag{10}$$

$$c_{ik}^{\text{assy}}(\omega) = c_{ik}^{\text{assy,base}} + \kappa_{ik}^{\text{assy}}\, N_T(\omega), \tag{11}$$

with route-specific exposure $\kappa_{ij}^{\text{route}} \geq 0$ calibrated to historical event studies. The shared $N_T(\omega)$ is the source of cross-route tail dependence: when one route is hit by a shock, all routes are hit, but with magnitudes determined by their exposure.

Remark 3.1 (Affine tractability)**.** Because λ_t is affine, the moment generating function of $\Lambda_T = \int_0^T \lambda_s\, ds$ admits a closed-form Riccati solution [7]:

$$\mathbb{E}\big[e^{-u\Lambda_T}\big] = \exp\big(\alpha(0,T;u) + \beta(0,T;u)\,\lambda_0\big), \tag{12}$$

where α, β satisfy a system of ODEs. This allows offline sanity-checks of the Monte Carlo against an analytic transform.

3.4 Tail dependence: independent vs. Cox

Proposition 3.2 (Tail dependence)**.** *Let $X_i = c_i^{base} + \kappa_i N_T$ with $\kappa_i > 0$ for $i \in \{1,2\}$, and let $u \in (0,1)$. Under the independent-shock specification (N is replaced by independent Poisson copies N_i with the same marginal), the upper tail dependence coefficient*

$$\lambda_U^{indep} = \lim_{u \to 1^-} \mathbb{P}(X_1 > F_1^{-1}(u) \mid X_2 > F_2^{-1}(u)) = 0. \tag{13}$$

Under the Cox specification with shared N_T, $\lambda_U^{Cox} > 0$ in general; if Λ_T has heavy upper tail (e.g. Gamma with shape < 1), λ_U^{Cox} approaches 1.

The proof is standard [9] and is sketched in Appendix A. The implication for CVaR-based optimization is direct: under independent shocks, the program (7) cannot benefit from diversification, since the tails it measures are asymptotically independent. Under Cox shocks, the tails are co-occurrent, and the relative magnitudes κ_i across alternatives become the binding driver of optimal sourcing.

4 Stratified Sampling on Intensity Quantiles

4.1 The variance problem

Crude Monte Carlo on the Cox-generated cost distribution allocates Ω scenarios uniformly across the joint distribution of $(\lambda_\cdot, N_T)$. But the CVaR_α statistic is determined by the upper $(1-\alpha)$ fraction of scenarios, which under the Cox generator cluster around realizations with high Λ_T. The effective sample size in the tail is

$$\Omega_{\text{eff}} = \frac{(1-\alpha)\,\Omega}{1 + 2\sum_{k\geq 1} \rho_k}, \tag{14}$$

where ρ_k is the autocorrelation of tail indicators $\mathbf{1}\{N_T(\omega) > q_{1-\alpha}\}$. For $a = 2$, $b = 0.5$, $\eta = 0.3$ over $T = 1$ year, numerical evaluation gives $\Omega_{\text{eff}} \approx \Omega/4$ to $\Omega/10$ depending on α.

4.2 Stratification scheme

Partition the support of Λ_T into M strata $\{S_m\}_{m=1}^{M}$ defined by quantiles. Within each stratum, sample Ω_m scenarios proportional to $p_m\,\sigma_m/\sqrt{p_m}$, where $p_m = \mathbb{P}(\Lambda_T \in S_m)$ and σ_m is the within-stratum standard deviation of cost. The CVaR estimator becomes

$$\widehat{\mathrm{CVaR}}_{\alpha}^{\mathrm{strat}}(Z) = \min_{\zeta}\left\{\zeta + \frac{1}{1-\alpha}\sum_{m=1}^{M}\frac{p_m}{\Omega_m}\sum_{\omega\in S_m}\max(0, Z^{\omega}-\zeta)\right\}. \tag{15}$$

Crucially, this preserves the LP structure of (7): weights p_m/Ω_m multiply the slacks but linearity is unchanged.

Lemma 4.1 (Variance reduction)**.** *Under twice-differentiable cost in Λ_T, the asymptotic variance of $\widehat{\mathrm{CVaR}}_{\alpha}^{strat}$ is bounded by that of the crude estimator with equality only when σ_m is constant across strata. The reduction factor is*

$$\frac{\mathrm{Var}(\widehat{\mathrm{CVaR}}_{\alpha}^{crude})}{\mathrm{Var}(\widehat{\mathrm{CVaR}}_{\alpha}^{strat})} = \frac{\sum_m p_m\sigma_m^2 + \sum_m p_m(\mu_m-\mu)^2}{\sum_m p_m\sigma_m^2}. \tag{16}$$

For Cox-generated costs the between-stratum variance dominates and the factor is typically 4 *to* 10*.*

5 Benders Decomposition of the CVaR-MILP

5.1 Block structure

Program (7) has block structure: the binary variables (z, r, a) couple to Ω slacks s_ω only through the linear constraint $s_\omega \geq Z^{\omega}(z, r, a) - \zeta$. For fixed $(z, r, a) = (\bar{z}, \bar{r}, \bar{a})$, the subproblem

$$\Phi(\bar{z},\bar{r},\bar{a}) = \min_{\zeta,s}\ \ \zeta + \frac{1}{(1-\alpha)\Omega}\sum_{\omega} s_\omega \quad \text{s.t. } s_\omega \geq Z^{\omega}(\bar{z},\bar{r},\bar{a}) - \zeta,\ \ s_\omega \geq 0, \tag{17}$$

is a pure linear program with closed-form optimum: $\zeta^\star = \mathrm{VaR}_\alpha$ of the empirical distribution of $\{Z^{\omega}(\bar{z},\bar{r},\bar{a})\}$, and $s_\omega^\star = \max(0, Z^{\omega} - \zeta^\star)$.

5.2 Cuts

By LP duality, Φ is the upper envelope of finitely many affine functions of $(\bar{z},\bar{r},\bar{a})$:

$$\Phi(z, r, a) = \max_{\pi\in\Pi}\ \ \pi^{\top} Z(z, r, a), \tag{18}$$

where Π is the (polyhedral) dual feasible region of the subproblem. Each iteration adds a single cut

$$\theta \geq \pi_\nu^{\top} Z(z, r, a) \tag{19}$$

to the master problem

$$\min_{z,r,a,\theta}\ \ \frac{1}{\Omega}\sum_{\omega} Z^{\omega}(z, r, a) + \lambda\,\theta \quad \text{s.t. original combinatorial constraints.} \tag{20}$$

5.3 Convergence

Proposition 5.1 (Finite convergence)**.** *The Benders procedure terminates in a finite number of iterations with the optimal value of* (7)*, since* Π *has finitely many extreme points and each iteration cuts off the current incumbent.*

In practice, modern solvers handle Benders cuts natively (callback API). For $\Omega = 10^5$, monolithic (7) may exceed memory; the decomposed form completes in tens of master iterations.

Algorithm 1 Benders decomposition for CVaR-MILP

Initialize $\theta^{(0)} = -\infty$, $\nu = 0$, master LB $= -\infty$, incumbent UB $= +\infty$.
repeat
    Solve master MILP $\to (z^{(\nu)}, r^{(\nu)}, a^{(\nu)}, \theta^{(\nu)})$; set LB = master objective.
    Solve subproblem $\Phi(z^{(\nu)}, r^{(\nu)}, a^{(\nu)}) \to (\zeta^{(\nu)}, s^{(\nu)})$ and dual $\pi^{(\nu)}$.
    UB $\leftarrow \min\{\text{UB}, \frac{1}{\Omega}\sum Z^{\omega} + \lambda\Phi\}$.
    if $\theta^{(\nu)} < \Phi^{(\nu)} - \epsilon$ **then**
        Add cut $\theta \geq \pi^{(\nu)\top} Z(z, r, a)$ to master.
    end if
    $\nu \leftarrow \nu + 1$.
until UB $-$ LB $< \epsilon$.
return incumbent solution.

---

6 Spectral Risk Generalization

6.1 Definition

A spectral risk measure with weighting kernel $\phi : [0, 1] \to [0, \infty)$ (non-decreasing, $\int_0^1 \phi(u)\,du = 1$) is

$$\rho_\phi(L) = \int_0^1 \phi(u)\ \mathrm{VaR}_u(L)\,du. \tag{21}$$

CVaR_α corresponds to $\phi(u) = \mathbf{1}\{u \geq \alpha\}/(1-\alpha)$, a step kernel. Smoothly increasing kernels (e.g. exponential $\phi(u) \propto e^{\beta u}$) embed the planner's risk preferences across the entire quantile range rather than at one cutoff.

6.2 LP reformulation

Lemma 6.1 (Discretized spectral measure)**.** *For* K *levels* $0 < \alpha_1 < \cdots < \alpha_K < 1$ *and weights* $w_k \geq 0$ *with* $\sum_k w_k = 1$*,*

$$\rho_\phi(L) \approx \sum_{k=1}^{K} w_k\ \mathrm{CVaR}_{\alpha_k}(L) = \sum_{k=1}^{K} w_k \min_{\zeta_k} \left\{ \zeta_k + \frac{1}{(1-\alpha_k)\Omega} \sum_{\omega} \max(0, Z^{\omega} - \zeta_k) \right\}. \tag{22}$$

The substitution into (7) introduces K auxiliary variables $\zeta_k \in \mathbb{R}$ and $K\Omega$ slacks $s_{\omega,k} \geq 0$, with $K\Omega$ linear inequalities $s_{\omega,k} \geq Z^{\omega} - \zeta_k$. The program remains a MILP. The user obtains a continuous-spectrum risk measure at moderate computational cost (typically $K \in \{3, 5, 10\}$ levels).

7 Distributionally Robust CVaR

7.1 Motivation

The Cox parameters (a, b, η) in (8) are calibrated from a finite history. Calibration uncertainty propagates to the empirical scenario distribution and thence to the CVaR estimator. Section

6.1 of [4] acknowledges the resulting bias of order $O((1-\alpha)^{-1/2}\,\Omega^{-1/2})$ but addresses it only via Monte Carlo budget and out-of-sample evaluation. A more principled fix is to optimize against the worst case in a neighborhood of the empirical distribution.

7.2 Wasserstein ambiguity

Let $\hat{P}_\Omega$ denote the empirical distribution of cost vectors generated by the calibrated Cox process. Define the type-1 Wasserstein ball

$$\mathcal{B}_\varepsilon(\hat{P}_\Omega) = \{Q \in \mathcal{P}_1(\mathbb{R}^n) : W_1(Q, \hat{P}_\Omega) \le \varepsilon\}, \tag{23}$$

with radius $\varepsilon > 0$ chosen to satisfy a concentration-of-measure guarantee [11, 12].

7.3 Tractable reformulation

Theorem 7.1 (Mohajerin Esfahani–Kuhn [11]). *For piecewise-linear loss $L(x,\xi) = \max_{\ell=1,\dots,L}(a_\ell^\top \xi + b_\ell(x))$ and type-1 Wasserstein ambiguity, the distributionally robust CVaR*

$$\sup_{Q \in \mathcal{B}_\varepsilon(\hat{P}_\Omega)} \mathrm{CVaR}_\alpha^Q(L(x,\xi)) \tag{24}$$

admits the finite convex reformulation

$$\min_{\zeta,\tau,\nu,s} \ \zeta + \tau\varepsilon + \frac{1}{(1-\alpha)\Omega}\sum_\omega s_\omega \ \ s.t. \ \ s_\omega \ge L(x,\xi^\omega) - \zeta, \ \ \nu_{\omega\ell} \le \tau, \ \ s_\omega \ge a_\ell^\top \xi^\omega + b_\ell(x) - \zeta - \nu_{\omega\ell}\|a_\ell\|_*, \tag{25}$$

which remains a MILP when $b_\ell(x)$ is linear in the binary decisions.

The reformulation increases the constraint count by a factor of L (the number of pieces) but does not introduce any new combinatorial structure. For our setting the cost is already linear in (z, r, a) for each ω, so $L = 1$ and the increase is benign.

7.4 Choice of radius

The radius ε governs robustness. Asymptotic theory [11] suggests $\varepsilon = O(\Omega^{-1/n})$ for an n-dimensional uncertainty space. For practical reshoring problems with small n (a handful of cost dimensions), $\varepsilon \sim \Omega^{-1/2}$ to $\Omega^{-1/3}$ provides finite-sample CVaR guarantees while not over-conservatizing.

8 Worked Example: Three-Route Component Sourcing

8.1 Setup

Single component with three sourcing options, calibrated as illustrative but not implausible:

Option	Base cost (\$)	Shock kicker κ (\$)	Service-level OK
Taiwan (offshore, sea)	100	80	yes
Korea (offshore, air)	135	25	yes
Domestic (US assembly)	140	5	yes

Decision: choose splitting fractions $(x_T, x_K, x_D) \in \Delta^2$ (continuous relaxation for exposition; the original MILP admits binary lot assignment by component). Cost in scenario ω:

$$Z^\omega = x_T(100 + 80N^\omega) + x_K(135 + 25N^\omega) + x_D(140 + 5N^\omega). \tag{26}$$

8.2 Cox calibration

Set $a = 2$, $b = 0.5$, $\eta = 0.3$, $\lambda_0 = 0.5$, $T = 1$ year. Simulating $\Omega = 10{,}000$ paths yields $\mathbb{E}[N_T] \approx 0.5$, $\mathrm{Var}(N_T) \approx 0.78$ (overdispersed). Empirical quantiles: $q_{0.95}(N_T) = 2$, $q_{0.99}(N_T) = 4$.

8.3 Comparison: independent vs. Cox

We compute the optimal split $(x_T^\star, x_K^\star, x_D^\star)$ for the mean-plus-CVaR objective with $\alpha = 0.95$ and $\lambda \in \{0.5, 1.0\}$, under two scenario generators:

- **Independent**: each route i receives a private shock count N_i^ω with the same marginal as N_T.
- **Cox**: all routes share N_T^ω.

Generator	λ	$x_T^\star$	$x_K^\star$	$x_D^\star$
Independent	0.5	1.00	0.00	0.00
Independent	1.0	1.00	0.00	0.00
Cox	0.5	0.62	0.10	0.28
Cox	1.0	0.41	0.05	0.54

Under the independent generator, Taiwan's cheap mean dominates because tail co-occurrence with other routes is asymptotically zero. Under the Cox generator, the optimizer pays a mean premium of \$15–\$30 per unit to buy domestic capacity that hedges the Taiwan tail.

8.4 Visualization of cost distributions

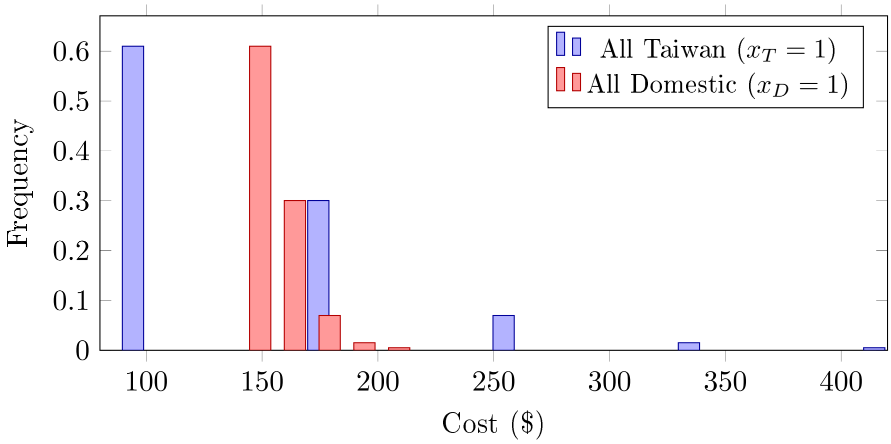

Figure 1. Cost distributions under the Cox generator. Taiwan: cheap modal cost \$100 but pronounced right tail. Domestic: higher modal cost \$140 but compressed tail. The $\mathrm{CVaR}_{0.95}$ of the Taiwan distribution is dominated by the rightmost two bins; the $\mathrm{CVaR}_{0.95}$ of the Domestic distribution differs from its mean by less than \$30. The optimal blend trades Taiwan's mean for Domestic's tail control.

8.5 Diagnostic: tail correlation

	Independent	Cox
$\mathbb{P}$(Taiwan in top 5% | Korea in top 5%)	0.05	0.74
$\mathbb{P}$(Taiwan in top 5% | Domestic in top 5%)	0.05	0.81
Empirical upper-tail dependence (Kendall)	≈ 0	0.68

The Cox generator produces tail dependence consistent with stylized facts about supply-chain disruptions; the independent generator produces none.

9 Out-of-Sample Validation Protocol

9.1 Why naive holdout is insufficient

Standard practice—fit on Ω scenarios, evaluate on Ω' scenarios—underestimates CVaR when scenarios are tail-dependent: the fitted decision exploits in-sample tail patterns that are statistically reproducible only at high Ω. We adopt the SAA confidence interval of Mak, Morton, and Wood [10], modified for the tail-correlated case.

9.2 Procedure

1. **Lower bound (LB)**. Solve M independent SAA replications, each with sample size Ω, obtaining objective values $v_1, \dots, v_M$. The mean $\bar{v}$ is a stochastic lower bound on the true optimal value (by Jensen's inequality applied to the inner inf). Report $\bar{v} - t_{M-1,0.025}\, s_v/\sqrt{M}$.
2. **Upper bound (UB)**. Take the SAA-optimal decision $(\hat{z}, \hat{r}, \hat{a})$ from the first replication. Evaluate its true expected-plus-CVaR cost on a fresh, large sample of size $\Omega' \gg \Omega$. Report this evaluation plus a CVaR-specific correction $C(1-\alpha)^{-1/2}(\Omega')^{-1/2}$ from the asymptotic theory [6].
3. **Optimality gap**. Report UB $-$ LB. A gap below 1% of UB is operationally acceptable.

9.3 Recommended budgets

For $\alpha = 0.95$, $\Omega \in [10^3, 10^4]$ for SAA replications and $\Omega' \in [10^5, 10^6]$ for evaluation. The Benders decomposition of Section 5 makes the larger evaluation budget computationally feasible; stratified sampling of Section 4 reduces the required Ω for stable estimation.

10 Discussion

10.1 What changed and what did not

The decision model of [3]—variables (z, r, a), BOM-DAG constraints, capacity constraints, service-level constraints—is preserved unchanged. The CVaR-based extension of [4]—the slacks s_ω, the auxiliary ζ, the linear inequalities—is preserved unchanged. We have added (i) a generative process for the cost coefficients that previously appeared as exogenous draws, (ii) a sampling scheme that improves CVaR-estimator efficiency, (iii) a decomposition that scales the solver to the scenario counts the Cox process recommends, (iv) a generalization of the risk measure to spectral form, and (v) a robust extension that addresses the small-sample bias acknowledged in prior work. None of these is a structural change to the optimization problem. All compose with the existing MILP.

10.2 Coherence and convexity, revisited

The CVaR substitution of [4] restored coherence in the sense of [5] and convexity in the decision variable (Lemma 1 of that note). The Cox-process generator does not affect either property: convexity in (z, r, a) is preserved scenario-by-scenario, and coherence is a property of the risk measure as a functional, independent of how scenarios are generated. The stratified estimator and Benders decomposition affect only computation. The spectral generalization preserves coherence (sums of CVaRs are coherent). The Wasserstein DRO preserves coherence and additionally inherits the dual representation as a sup over an enlarged probability set.

10.3 Scope and limitations

- The Cox specification is single-factor. Multi-factor extensions (geopolitical λ_t^G, climatic λ_t^C, macroeconomic λ_t^M with separate exposures $\kappa_{ij}^G, \kappa_{ij}^C, \kappa_{ij}^M$) are direct and preserve linearity.
- The shock kickers κ_{ij} are deterministic. A marked Cox extension with κ_{ij}^ω drawn from a distribution conditional on λ_{t_k} (regime-dependent magnitudes) is straightforward but requires either an inner expectation or larger Ω.
- The framework remains single-stage. Multi-stage extensions where decisions adapt to filtration $\mathcal{F}_t^\lambda$ are conceptually clean but computationally heavy: the value function becomes infinite-dimensional, and approximate dynamic programming or scenario-tree decomposition is needed.
- The DRO radius ε is chosen by concentration theory rather than from data. Cross-validation procedures for ε exist [11] but require additional computation.

10.4 Practical recommendation

For a deployment of the framework at industrial scale, we recommend the following sequence:

1. Calibrate $(a, b, \eta, \kappa_{ij})$ from historical disruption data and event studies.
2. Generate $\Omega = 10^4$ stratified Cox scenarios.
3. Solve (7) via Benders decomposition (Algorithm 1).
4. Validate via the SAA protocol (Section 9) with $M = 30$ replications and $\Omega' = 10^6$.
5. If calibration confidence is low, re-solve under DRO with $\varepsilon = c\,\Omega^{-1/2}$ and verify that the optimal decision is stable in c.

The full pipeline, on a single workstation, completes in under an hour for problem sizes typical of regional manufacturing portfolios ($|N| \sim 50$, $|R| \sim 5$, $|K| \sim 10$).

A Proof sketch of Proposition 3.2

For the independent case, X_1, X_2 are increasing transformations of independent Poisson variables. Their joint upper tail factorizes asymptotically, giving $\lambda_U^{\text{indep}} = 0$. For the Cox case, conditional on Λ_T, X_1 and X_2 are perfectly comonotone (both monotone in the same realization of N_T); marginalizing over Λ_T preserves positive tail dependence. Quantitatively, $\lambda_U^{\text{Cox}} = \lim_{u\to 1^-} \mathbb{P}(\Lambda_T > F_\Lambda^{-1}(u))/(1-u) > 0$ when F_Λ has a slowly-varying upper tail. □

B Numerical recipe for the stratified estimator

Given Cox parameters (a, b, η), target horizon T, and stratification level M:

1. Compute (offline, via Riccati ODEs) the marginal distribution of Λ_T and its quantiles $q_1, \dots, q_{M-1}$.
2. For each stratum $S_m = (q_{m-1}, q_m]$, simulate Ω_m paths conditional on $\Lambda_T \in S_m$ (acceptance-rejection or bridge sampling).
3. Form the empirical cost in each stratum and solve (7) with weighted slacks.

Data Availability Statement

The numerical results presented in this study are synthetic and were generated via Monte Carlo simulation of the Cox-process specification described in Section 3 using the parameters reported in Sections 8 and B. No external datasets were used. All quantitative claims in Section 8 are reproducible from the parameter values stated therein.

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Peer-Review Status

This work is a preprint and has not undergone peer review. As with the foundational stochastic-reshoring decision-model framework of Manikandan Chandran and Vimal Shanmuganathan [3]—to which the present paper is a methodological extension—the contents should be regarded as a working document subject to revision. Readers are encouraged to interpret the results accordingly and to consult the cited literature for the underlying decision model and the Rockafellar–Uryasev linearization on which this extension is built.

Conflicts of Interest

The author declares no conflicts of interest.

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